

AI for Bharat Hackathon

Powered by **aws**



Team Name : ByteCoke

Team Leader Name : Tanisha Mathur

Problem Statement : AI-powered welfare eligibility for India's underserved.

NYAAYA.ai

Conversational AI for Welfare



Voice-First Hinglish AI

Understands context, asks smart questions



Zero Hallucination

Widow Pension • Disability • NREGA



Week-by-Week Strategy

Ranked by benefit impact

65M+ Citizens Unaware → **NYAAYA.ai**
Solves It in Minutes



Why AI?

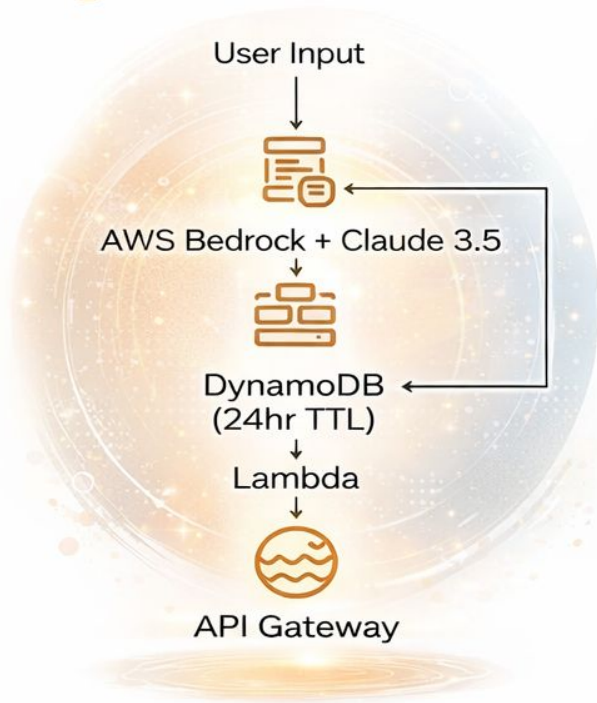


- Complex Eligibility
- Conversational Nuance
- Scaling to 65M Citizens



- Scaling to 65M Citizens

Powered by AWS



AI Layer Benefits

- <2s Response Time {
- Hinglish Support {
- 100% Success Rate (73/73 tests)



- <2s Response Time
- Hinglish Support
- 100% Success Rate (73/73 tests)

FEATURES THAT POWER NYAAYA.AI

Built for India's Citizens



Multilingual

- Hinglish conversational AI interface
- Accepts voice or text input
- Handles informal phrasing used by citizens

Example: "Widow pension ke liye kya chahiye?"



Rule Engine

- Hardcoded government scheme eligibility criteria
- No AI hallucinations or guessing
- Transparent and explainable decision logic

Ensures accurate eligibility determination



Strategy Optimizer

- Evaluates multiple welfare schemes simultaneously
- Prioritizes by financial benefit and approval likelihood
- Generates step-by-step application roadmap

Example: Pension – Health – Employment



Low-Bandwidth Optimized

- Designed for 2G / rural connectivity
- Average response time <2 seconds
- Minimal data usage

Works reliably even in low-internet regions



Zero-Failure Fallback

- Fully rule-based backup system
- Works even if AI layer fails
- Offline eligibility checks possible

Citizens always receive an answer



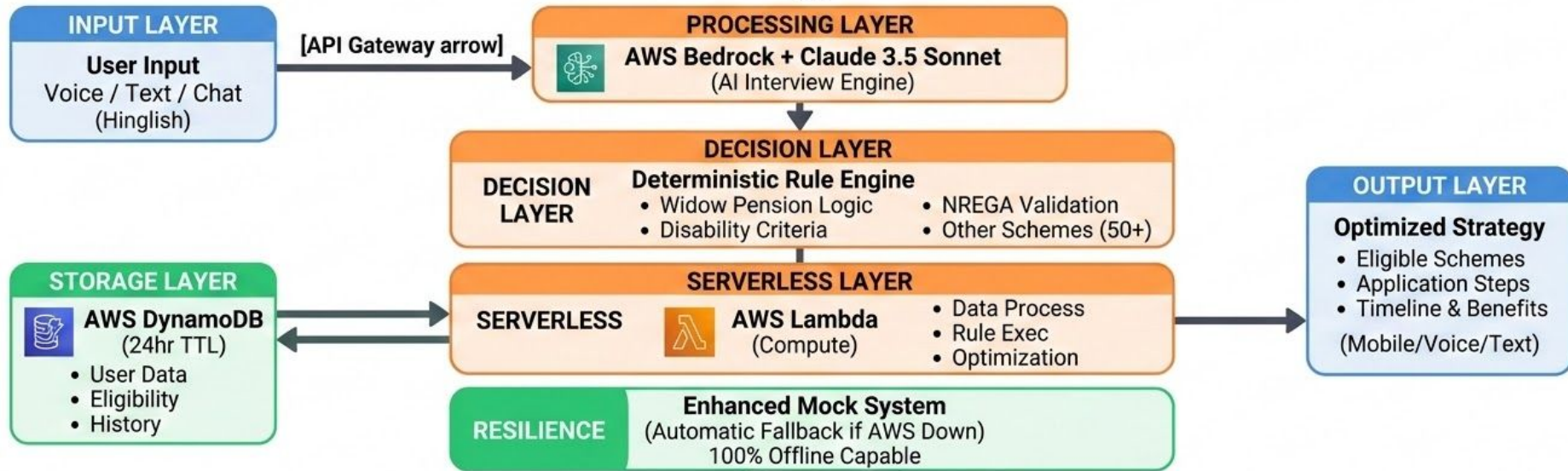
Secure & Private

- No sensitive data stored permanently
- Encrypted processing pipeline
- Citizen data protected by design

How It Works



Technology Stack



User Interface & Experience Design

How citizens interact with NYAAYA.ai

Natural Hinglish conversation

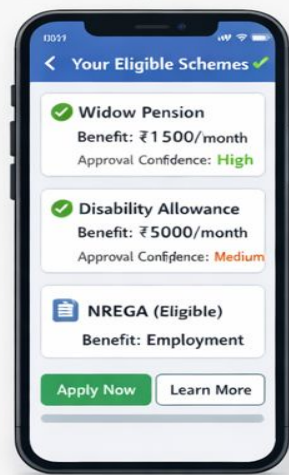


Real-time eligibility calculation



Ranked by benefit impact

One-click application



Start Interview

Answer Questions

Apply to Schemes

Start Interview

Ask your question

Answer Questions

Fill in your details

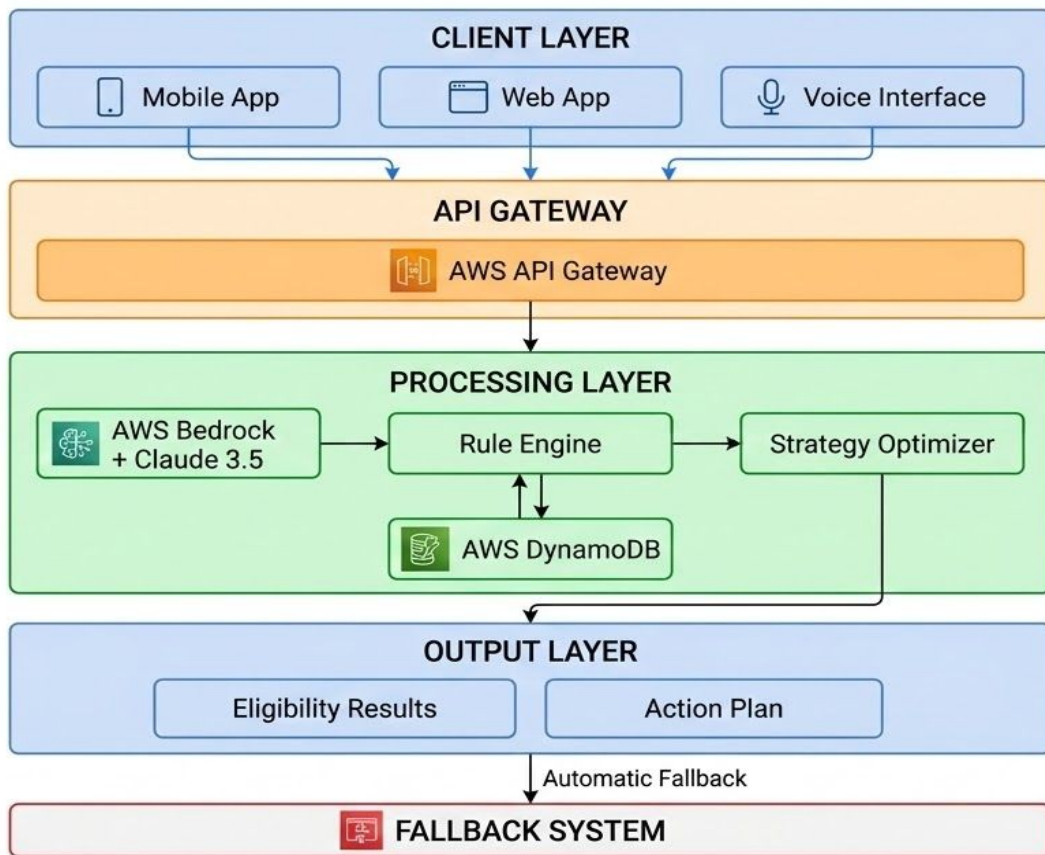
Get Results

View eligible schemes

Apply to Schemes

Submit your application

SIMPLE TECHNICAL ARCHITECTURE DIAGRAM FOR NYAAYA.ai



\$3.35/month



Operational Cost

✓ **\$0.0000034**
per citizen (at scale)

✓ **Scales to 65M+ citizens**
at **SAME** cost

✓ **97% cheaper than**
traditional approach

NYAAYA.ai

Traditional

\$3.35/month

\$500K-1M/month

<2 minutes

2-3 days

AI & LLM



AWS Bedrock
Claude 3.5

BACKEND



Python 3.12
FastAPI
AWS Lambda

DATABASE



DynamoDB
24-hr TTL

INFRASTRUCTURE



AWS Services
99.9% Uptime

BUSINESS LOGIC



Rule Engine
50+ Schemes
Zero Hallucinations

RESILIENCE



Offline-First
Mock System
100% Available

FRONTEND



React Native
React.js
Hinglish UI

MONITORING



CloudWatch
<2s Response
24/7 Available

←  Nyaaya.ai

Main ek kisan hoon

Aap kaun se state aur district se hain?

rajasthan mein. jaipur district main.

Aapki age aur Income kitni hai?

27 aur 100000

Aap shadi shudha hain? Ya single hain?

ha shadi shudha hoon

Aap gaon mein rahte hain ya shehar mein?

Type or speak in Hindi/English...

←  Nyaaya.ai

Confidence 93%

Eligibility Summary

1 of 6 schemes match your profile.

1	₹500	₹6,000
Schemes Eligible	Per Month	First Year Total

Eligible Schemes (1)

PM-KISAN Samman Nidhi

Eligible

You meet all eligibility criteria: registered as farmer, income ₹100000 is below ₹2,00,000 threshold, rural resident, has Aadhaar card

₹500/mo 6 weeks

Confidence: 93% Approval rate: 92%

Required Documents

aadhaar land ownership record bank passbook kisan registration

←  Nyaaya.ai

Priya S.

30-39 yrs
Lucknow, Uttar Pradesh
PM Ujjwala Yojana (Free LPG)
4 weeks 94% approval

Tips

- Apply at nearest LPG distributor with BPL card
- Free connection + first refill included
- Subsidy comes directly to bank account via DBT

Savitri D.

40-49 yrs
Pune, Maharashtra
Widow Pension Scheme
8 weeks 91% approval

Blockers faced

- Death certificate had spelling mismatch with Aadhaar name

Tips

- Get name corrected on death certificate BEFORE applying
- Carry 3 photocopies of every document
- Apply through Gram Sevak — faster than CSC

Raju M.

30-39 yrs
Nagpur, Maharashtra
Disability Allowance
7 weeks 85% approval

Blockers faced

- Private hospital disability cert was rejected

Prototype Performance Report & Benchmarking

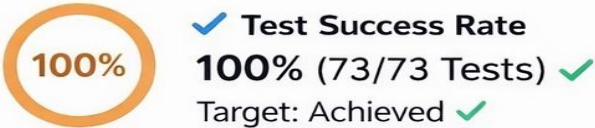
Production-ready system with exceptional performance metrics

CORE PERFORMANCE METRICS



Average response time per query, optimized for 2G networks

CORE PERFORMANCE METRICS



All unit tests passing, zero critical failures

CORE PERFORMANCE METRICS



Production-ready reliability with minimal downtime

ACCURACY & RELIABILITY METRICS



Deterministic rule engine ensures perfect accuracy, no AI hallucinations

ACCURACY & RELIABILITY METRICS



System handles enterprise-scale concurrent usage without degradation

BENCHMARKING COMPARISON

Response Time	Accuracy	Cost Efficiency	Scalability
NYAAYA.ai	<2 seconds (88% faster)	\$3.35/month	Unlimited (Same cost)
Alternative A	15 seconds	\$50K+/month	Limited (Costs increase)
Alternative B	20 seconds	\$100K+/month	Very Limited
NYAAYA.ai	\$35/month	\$50K+/month	\$100K+/month
Alternative A	15 seconds	\$100K+/month	Very Limited
Alternative B	20 seconds		

Additional Details & Future Development

Roadmap for scaling impact across India



PHASE 1

Q1-Q2 2026

- ✓ New schemes
- ✓ Mobile app
- ✓ State portals
- ✓ SMS support
- ✓ Better Hinglish

PHASE 2

Q3-Q4 2026-2027

- ✓ South Indian languages
- ✓ More schemes
- ✓ Analytics
- ✓ Success stories

PHASE 3

2027+

- ✓ All languages
- ✓ All states
- ✓ One-click
- ✓ International
- ✓ Full scale

Prototype Assets:

→ **GitHub Public Repository Link :**

<https://github.com/taneem77/AI-For-Bharat---Nyaaya-AI>

→ **Demo Video Link (Max: 3 Minutes) :**

https://drive.google.com/file/d/1Bnx7lc9hghfCSQSYEZMdNB6MepG8_SG/view?usp=sharing

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Thank You

